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Innovative Near-Surface Model Building via Multi-Domain SJI: Insights from Onshore US

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Summary

The increasing complexity of land-based hydrocarbon exploration necessitates the development of high-fidelity, high-resolution 3D seismic imaging techniques. In many onshore environments, geological features such as buried karsts, paleochannels, and anthropogenic fill zones introduce rapid lateral velocity variations that can significantly distort seismic wave propagation. These distortions often manifest as mispositioned reflectors, imaging artifacts, and a general loss of resolution, complicating both structural and stratigraphic interpretation.

This case study presents a depth reprocessing initiative aimed at addressing these challenges in a structurally complex onshore environment. The primary objectives were to suppress artificial imaging features introduced by near-surface heterogeneity and to enhance the visibility of subtle geological elements, including minor faulting and stratigraphic discontinuities that are critical for reservoir delineation.

To overcome these limitations, a simultaneous joint inversion (SJI) of refracted and surface waves was implemented. This advanced technique integrates complementary wavefield information to produce a high-resolution near-surface velocity model. By jointly inverting both wave types, the method leverages the sensitivity of refracted waves to deeper velocity structures and the sensitivity of surface waves to shallow layers. By integrating these datasets, the inversion produced a geologically consistent and high-fidelity near-surface velocity model.

The detailed near-surface model produced by SJI was then incorporated into the depth imaging workflow, significantly improving the accuracy of static corrections and the fidelity of the final seismic image. The results demonstrated a marked reduction in imaging artifacts, clearer delineation of fault systems, and enhanced visibility of subtle stratigraphic features that were previously obscured. This approach not only improved the overall quality of the seismic image but also increased interpreter confidence, supporting more informed decision-making in subsequent exploration and development phases.

Introduction

Accurate imaging of deeper subsurface targets critically depends on the mitigation of distortions introduced by near-surface heterogeneities. These distortions, if uncorrected, can significantly compromise the fidelity of seismic reflection data and the reliability of reservoir characterization. Therefore, constructing a detailed and geologically consistent near-surface velocity model is a foundational step in seismic depth imaging workflows.

Conventional approaches to near-surface velocity estimation typically rely on the analysis of refracted (diving) waves to derive compressional-wave (Vp) velocity models (Podvin et al., 1991). While effective under certain conditions, these methods are limited in their ability to resolve shallow velocity inversions and lateral heterogeneities. Alternatively, Rayleigh waves—when adequately sampled—offer a complementary source of information, particularly for estimating shear-wave (Vs) velocities (Strobbia et al., 2010).

Refraction tomography (RT) and surface-wave inversion (SWI) are based on fundamentally different physical principles and utilize distinct components of the seismic wavefield. These methods differ in terms of depth sensitivity, resolution characteristics, and susceptibility to acquisition-related limitations. Despite these differences, the physical properties they estimate—Vp and Vs—are strongly correlated. A joint inversion strategy that integrates RT and SWI can therefore exploit the complementary strengths of each method to produce a more accurate and robust near-surface velocity model.

Surface-wave analysis, known for its high lateral resolution and robustness, is particularly effective in resolving shallow subsurface complexities, including velocity inversions that are often undetectable in refraction data. It also compensates for limitations such as missing or noisy near-offset travel times. Conversely, refraction data extend the depth of investigation and enhance vertical resolution. The integration of these datasets facilitates a more accurate Vs-to-Vp conversion and yields a comprehensive, multi-parameter near-surface model.

Such a model not only improves the accuracy of near-surface corrections—such as static corrections essential for depth imaging—but also provides valuable insights for geological interpretation. In particular, the derived Vp/Vs ratio can inform lithological and geomechanical characterizations of the near surface. To demonstrate the practical utility of this integrated approach, we present a field case study from US land survey, highlighting its effectiveness in resolving complex near-surface conditions and enhancing subsurface imaging fidelity.

Data Description and Geophysical Objectives

The seismic dataset employed in this study comprises four distinct surveys, which are subsets of the Seitel and WesternGeco 3D land seismic campaigns. These surveys were conducted in a structurally complex onshore environment characterized by diverse acquisition parameters. While vibroseis sources were predominantly used, the dataset also includes contributions from dynamite and mini-vibroseis sources, introducing variability in source signatures and wavefield characteristics.

Figure 1 illustrates a representative example of the raw field data, highlighting the presence of strong Rayleigh waves, primarily dominated by the fundamental mode. These surface waves exhibit lateral velocity variations indicative of near-surface heterogeneity. The quality of the first arrivals is sufficient to support reliable picking, enabling robust inversion and velocity model building.

The study area features a geologically diverse set of formations, including the naturally fractured Austin Chalk, a key hydrocarbon reservoir; the organic-rich Eagle Ford Formation, which serves as both a source rock and a primary production target; the lithologically varied Wilcox Formation, important for hazard assessment; and the faulted Nacatoch Formation, which contributes to near-surface complexity and affects seismic wave behavior.

The primary geophysical objective of this study is to enhance seismic data quality through improved resolution, accurate imaging near salt bodies and beneath complex overburden, and preservation of amplitude and azimuthal fidelity. These enhancements are essential for hazard identification, accurate geo-steering, facture characterization and integration into high-fidelity 3D geological models.

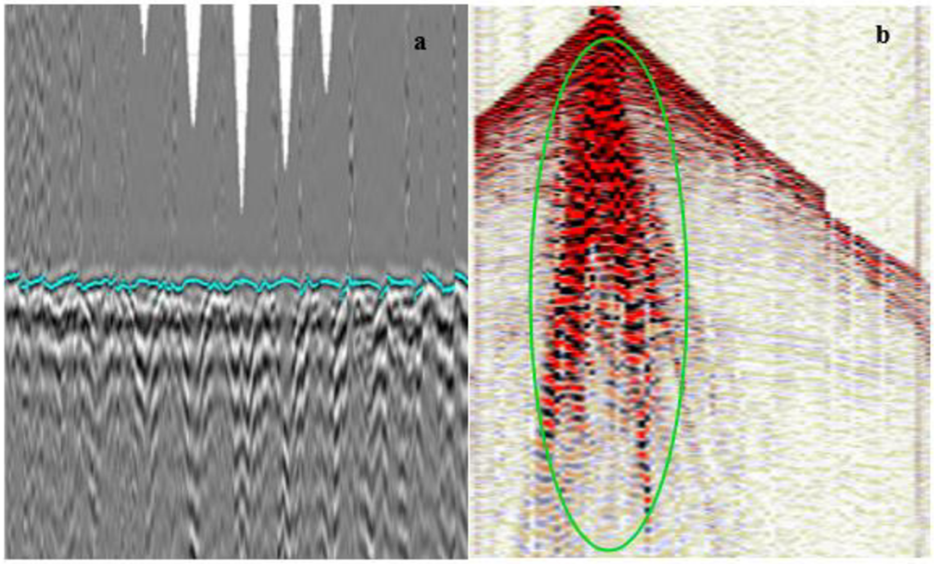

Figure 1—Examples of Field data (a) Refracted wave picks shown in cyan overlaid on shot gather with linear moveout applied; (b) Shot gather with surface waves highlighted in green.

To achieve these goals, it is imperative to construct a high-resolution near-surface velocity model that ties to available well data and mitigates imaging artifacts caused by rapid lateral and vertical velocity variations.

Methodology

The primary objective of this study is to characterize the near-surface section through the application of Simultaneous Joint Inversion (SJI) a Multiphysics inversion framework designed to jointly invert multiple seismic waveforms and, where applicable, non-seismic datasets. SJI leverages the complementary sensitivity of various geophysical measurements, including refracted and surface (Rayleigh) waves, as well as auxiliary data such as gravity and magnetic surveys, to construct a geologically consistent, high-resolution near-surface velocity model.

The term joint inversion refers to the integration of multiple geophysical measurements into a unified, multi-domain Earth model. Unlike cooperative joint inversion, which alternates between single-domain inversions and subsequently reconciles the results, simultaneous joint inversion integrates all measurements concurrently. This is achieved through the application of petrophysical and/or structural soft constraints, enabling robust coupling between datasets with differing sensitivities and resolution characteristics.

By simultaneously inverting distinct components of the seismic wavefield, SJI eliminates the need for a posteriori reconciliation between independently derived models. This approach enhances model consistency and resolution, particularly in complex near-surface environments where conventional methods may fail to resolve features such as shallow velocity inversions—commonly referred to as the "hidden layer" problem (Re, 2010; Speziali, 2017).

SJI offers a robust framework for integrating multiple seismic measurements through the application of petrophysical and/or structural soft constraints. This methodology enables the concurrent inversion and iterative updating of both compressional-wave (Vp) and shear-wave (Vs) velocity models, using refracted and surface wave data as input. The result is a high-resolution, multi-parameter, and self-consistent subsurface model that captures the spatial distribution of both Vp and Vs with improved accuracy.

By explicitly accounting for and modeling distinct components of the seismic wavefield, SJI circumvents the need for post-inversion reconciliation between independently derived models. This integrated approach not only enhances the internal consistency of the velocity models but also improves their geological plausibility, making them more suitable for subsequent applications such as depth imaging, static corrections, and near-surface characterization.

The SJI workflow (Figure 2) was initiated following the picking of first arrival times, which were used to assess acquisition geometry accuracy and to construct an initial compressional-wave velocity (Vp) model. This model was generated using the delay-time solution based on the intercept-time method. In parallel, surface wave analysis was conducted to extract the dispersion curve of the fundamental Rayleigh wave mode. From these dispersion curves, an initial shear-wave velocity (Vs) model was estimated by converting phase velocity to shear velocity and frequency to depth.

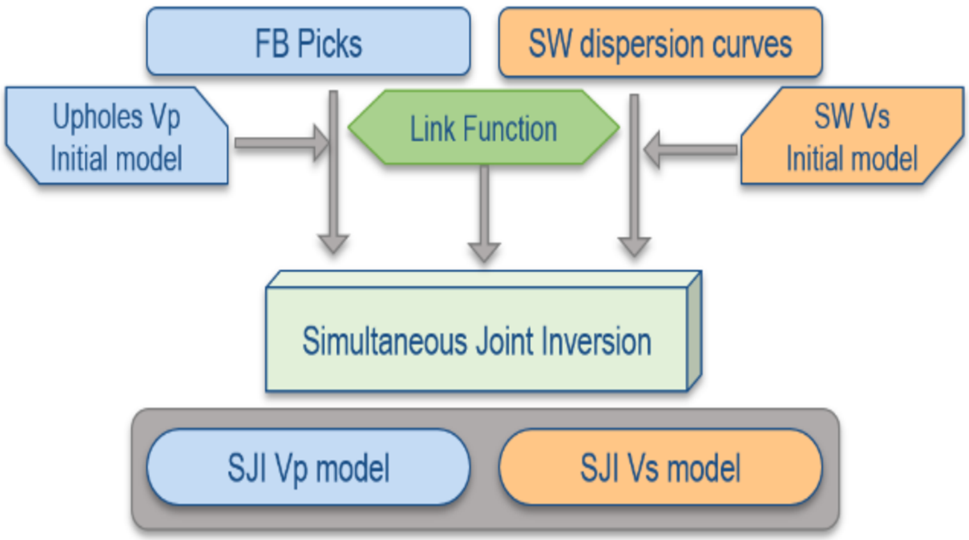

Figure 2—Flow chart of Simultaneous Joint Inversion (SJI) of refracted primary waves and surface waves, devised to maximize the information jointly extracted from the multi-wavefields data set while retaining the intrinsic resolution and depth of investigation of each of wavefields. The modelling resolution is progressively increased starting from single domain to multi-domain inversion.

Quality control (QC) of the Vp inversion was quantitatively evaluated by monitoring the convergence behavior between observed and forward-modeled seismic responses. Specifically, offset-dependent misfit analysis of first-break picks (Figure 3a) and frequency-dependent root-mean-square (RMS) misfit of Rayleigh wave dispersion curves for the Vs model demonstrated satisfactory convergence after five iterative updates.

To establish a physically meaningful relationship between the independently inverted Vp and Vs fields, a linking function was derived. This was achieved by cross-plotting high-confidence overlapping regions of the two models and fitting a best-fit regression curve. The resulting empirical relationship was then applied as a soft constraint in the subsequent SJI process.

The initial Vp and Vs models, along with the derived linking function, were incorporated into the SJI framework. Through multiple iterations, the inversion simultaneously refined both velocity fields, resulting in a high-resolution, geologically consistent, and self-consistent multi-parameter model.

Tracking of the objective function for all three components—Vp, Vs, and the linking constraint—alongside the aforementioned misfit metrics, indicated stable and reliable convergence behavior. Notably, the multi-domain SJI approach enhanced model stability and further reduced misfit values (Figure 3c) when compared to single-domain inversion results (Figure 3b). Additional qualitative assessments of the final model, as presented in the Results section, confirm improvements in geological plausibility and imaging fidelity.

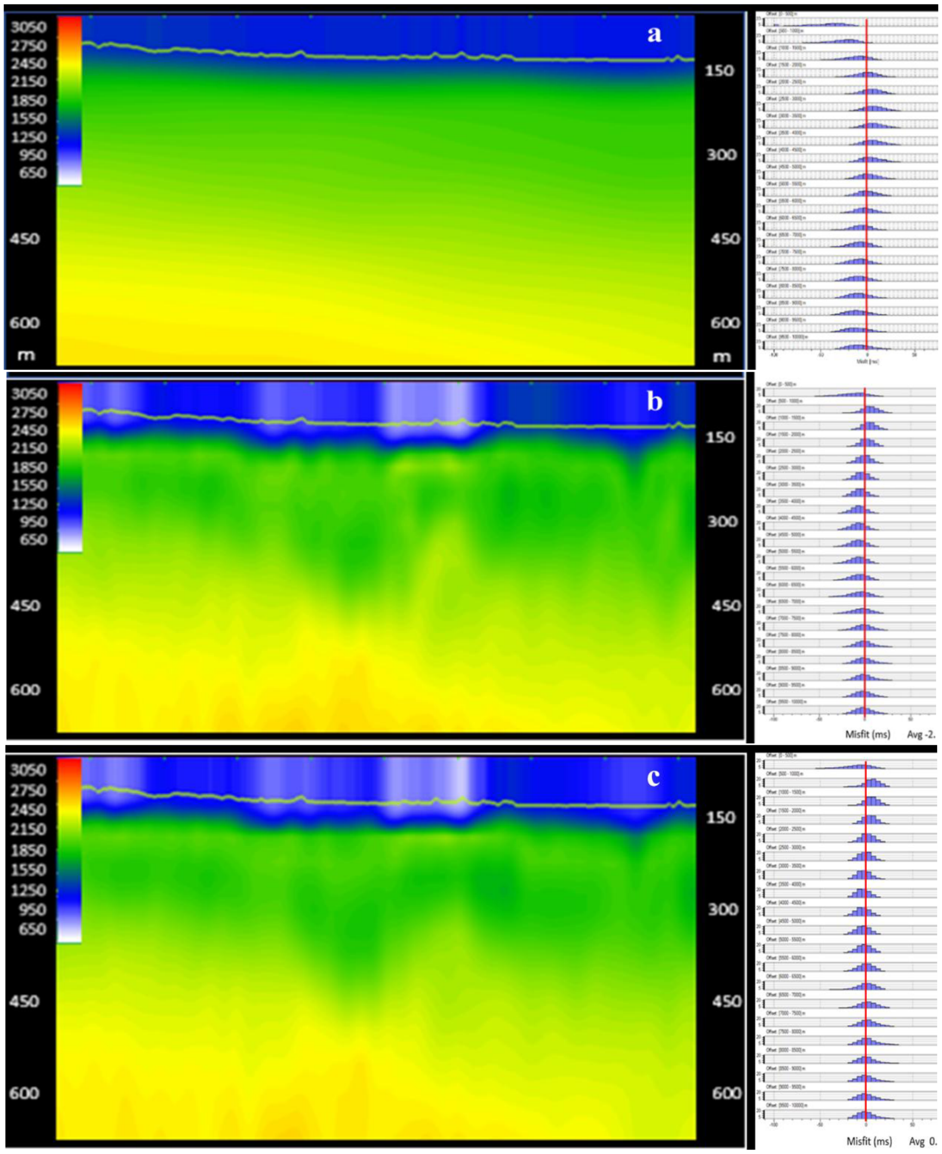

Figure 3—Vp velocity section and misfit histogram from (a) initial model, (b) from single domain inversion, and (c) from multi-domain simultaneous joint inversion.

The velocity model generated through Simultaneous Joint Inversion (SJI) effectively captures both the very shallow subsurface architecture and the complex heterogeneity of the near-surface zone, extending seamlessly into deeper refractors. This depth range overlaps with intervals typically refined using Common Image Point (CIP) tomography, thereby enhancing the continuity and consistency of the Earth model across multiple spatial scales.

Following its derivation, the multi-parameter velocity model—comprising both compressional-wave (Vp) and shear-wave (Vs) components—was integrated into the depth imaging workflow. It served as the near-surface portion of the full Earth model, replacing conventional statics-based corrections with a physically consistent velocity field. In addition to its role in depth imaging, the model was employed to compute both high-frequency and low-frequency static corrections for time-domain preprocessing.

This dual application underscores the versatility of the SJI-derived model and its contribution to improving seismic data quality. Specifically, it enhanced resolution, reduced imaging artifacts, and improved the accuracy of reflector positioning. The integration of a geologically consistent near-surface model also facilitated better amplitude preservation and improved azimuthal fidelity, supporting advanced interpretation tasks such as fracture characterization and reservoir delineation.

Results

The Simultaneous Joint Inversion (SJI) method demonstrated superior capability in resolving near-surface heterogeneity compared to conventional refraction-based approaches. This was quantitatively verified by computing static corrections from both models and evaluating their impact on seismic data quality. Statics derived from the SJI velocity model yielded improved structural continuity and enhanced stack coherency relative to those obtained from refraction tomography.

A shallow depth slice through the refraction based Vp solution (shown in Figure 4a) exhibits near surface complexity and heterogeneity. The same depth slice through the SJI derived Vp model is shown in Figure 4b. The relative magnitude of near surface heterogeneity is increased with the SJI model. The corresponding Vs model derived from SJI is shown in Figure 4c.

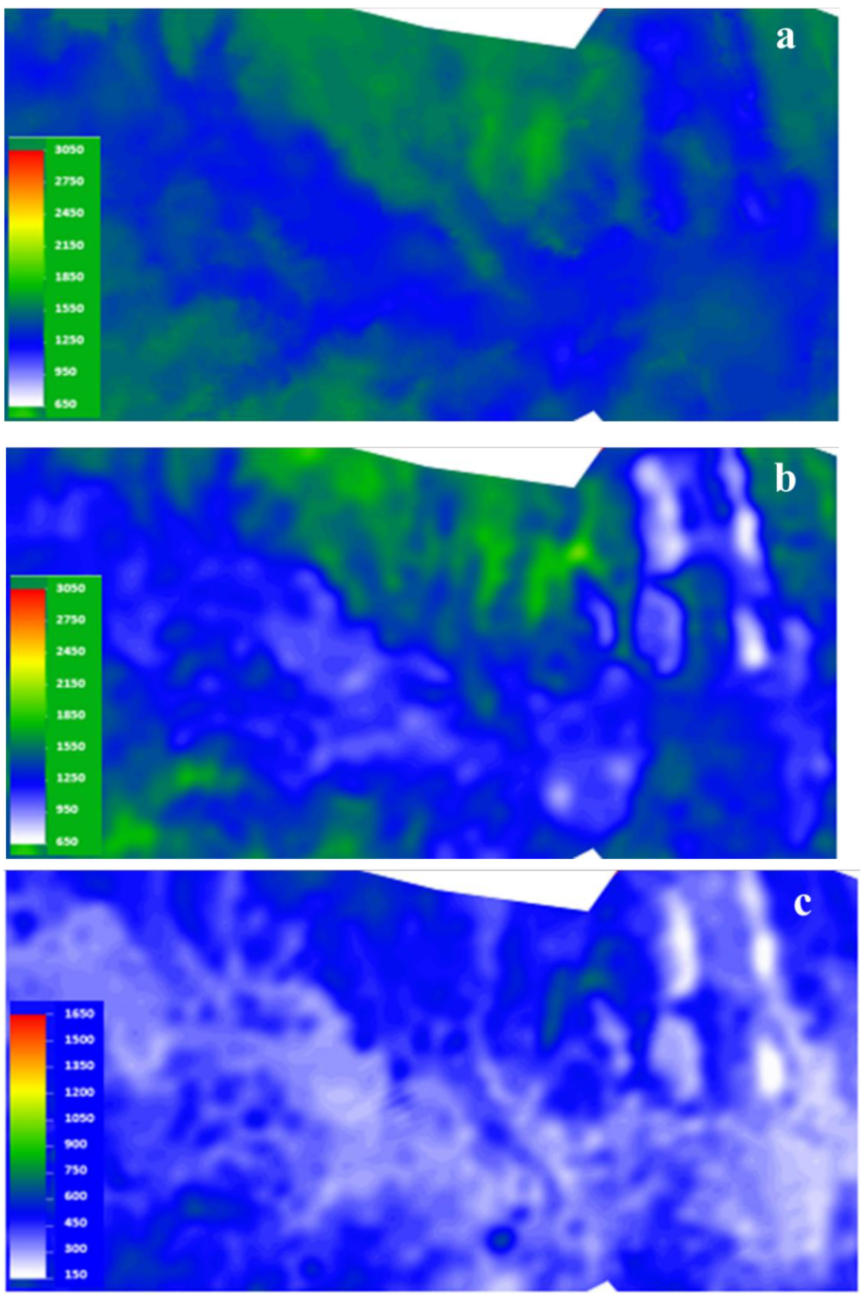

Figure 4—Depth slice of velocity fields at 360m below common datum: (a) Vp derived from refraction tomography, (b) Vp and (c) Vs derived from simultaneous joint inversion.

Static corrections computed from the SJI-derived velocity model were applied to the seismic dataset and compared against those obtained from a conventional refraction-based statics solution. The resulting stacked section using SJI statics (Figure 5b) exhibited significantly improved reflection coherency relative to the refraction-based stack (Figure 5a). This improvement highlights the enhanced capability of the SJI method in accurately capturing near-surface velocity variations.

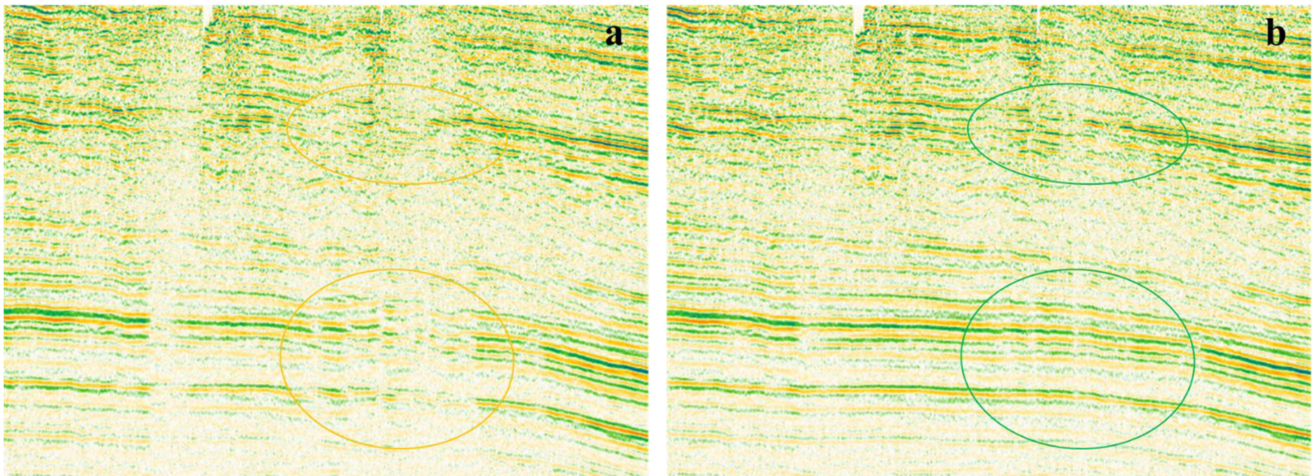

Figure 5—Premigration stack section examples after application of model statics derived from (a) refraction tomography, (b) SJI.

The improved alignment of seismic reflections, particularly in areas affected by acquisition noise, facilitated more effective noise attenuation during preprocessing. The enhanced signal coherency not only improved the interpretability of the seismic data but also enabled more accurate signal processing workflows. These results underscore the practical value of SJI in both imaging and preprocessing, contributing to higher-quality seismic products.

Imaging tests conducted across various geological settings further substantiated the superiority of the Seismic Joint Inversion (SJI)-derived model compared to conventional refraction-based velocity models. The SJI approach yielded structurally simpler and geologically consistent velocity fields, characterized by enhanced lateral continuity and amplitude stability. These improvements translated into seismic images with higher structural coherency and interpretability, particularly in complex near-surface environments where traditional methods often struggle with velocity heterogeneity and noise sensitivity.

To operationalize the advantages of the SJI model within the broader seismic imaging workflow, the near-surface velocity field obtained from SJI was seamlessly integrated into the initial depth imaging deep velocity model. This integration was performed through a model-merging strategy that preserved the high-resolution near-surface detail from SJI while maintaining compatibility with deeper velocity trends.

By incorporating the SJI-derived near-surface velocities, the imaging process was able to directly account for and correct near-surface distortions, thereby reducing the reliance on large-scale time-domain static corrections. This not only preserved the kinematic fidelity of the seismic wavefield but also ensured that the structural uplift observed in the migrated images was geologically meaningful rather than an artifact of post-processing.

Furthermore, the improved initial model facilitated more stable and geologically constrained updates during subsequent reflection-based velocity model building, particularly in reflection tomography. The enhanced starting model reduced inversion non-uniqueness and improved convergence behavior, ultimately leading to more accurate and reliable subsurface velocity models and depth images.

Following the model merge, six iterations of anisotropic Common Image Point (CIP) tomography were performed, incorporating well log information to further refine the velocity field. The inclusion of the detailed near-surface model from multi-domain SJI significantly improved the convergence and reliability of the tomography updates.

By simultaneously inverting refracted and surface wave measurements, the SJI methodology enabled a more comprehensive characterization of the seismic wavefield and a more accurate representation of the shallow subsurface. This integrated model facilitated improved preprocessing and Earth model construction, ultimately yielding a high-resolution seismic image with enhanced structural fidelity, clearer fault delineation, and superior well ties compared to legacy processing workflows (Figure 6).

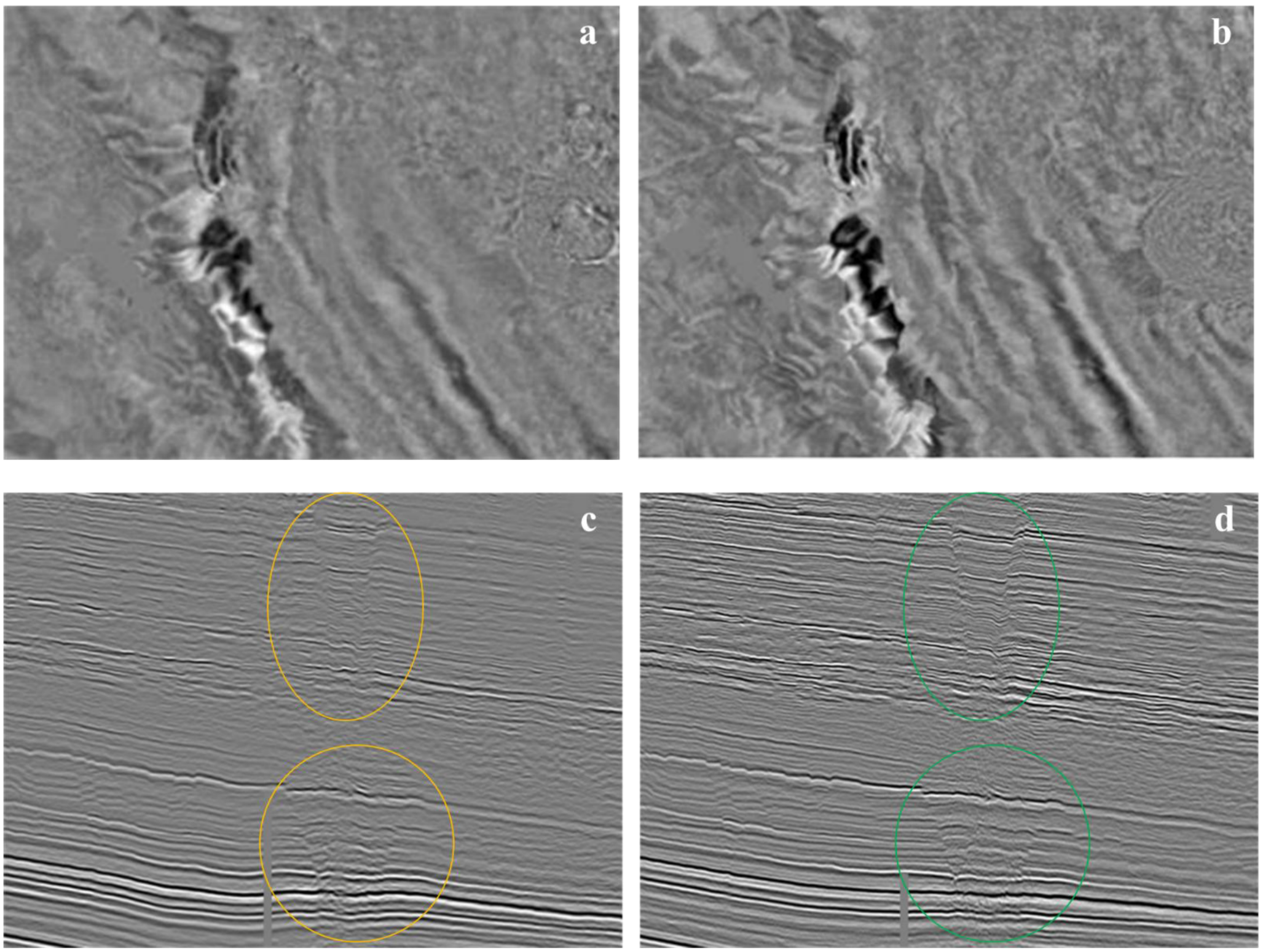

Figure 6—Kirchhoff pre-stack depth migrations: depth slice and profile from legacy processing (a, c), and from reprocessing effort (b, d).

Conclusions

This study has demonstrated that multi-domain Simultaneous Joint Inversion (SJI) represents a substantial advancement over traditional near-surface modeling methodologies. By incorporating additional components of the seismic wavefield—specifically, surface (Rayleigh) waves in conjunction with refracted arrivals—SJI facilitates the development of a high-resolution, geologically consistent near-surface model. The key advantages of this approach are as follows:

- **Enhanced Statics and Signal Coherency**: The SJI-derived model yields more precise static corrections, leading to improved signal alignment and stack coherency. These improvements contribute directly to more effective noise suppression and signal processing, particularly in complex land acquisition scenarios.

- **Comprehensive Velocity Model Construction**: Through the simultaneous inversion of compressional-wave (Vp) and shear-wave (Vs) velocity fields, SJI generates a multi-parameter model that provides a robust foundation for advanced imaging techniques, including reflection tomography and full-waveform inversion (FWI).
- **Robustness to Data Noise**: The methodology exhibits inherent resilience to measurement uncertainties and acquisition noise, which are prevalent in land seismic datasets. This robustness makes SJI particularly suitable for challenging near-surface environments where conventional methods often fall short.
- **Resolution of Subsurface Complexity**: Unlike refraction-based techniques, which are constrained by the "hidden layer" phenomenon, SJI is capable of resolving shallow velocity inversions. This capability enables a more complete and accurate characterization of the subsurface.
- **Flexibility and Expandability**: Although this case study employed only first-break picks and Rayleigh wave dispersion curves, the SJI framework is inherently adaptable. It can be extended to integrate additional geophysical measurements—such as gravimetric, electromagnetic, and uphole data—thereby further enhancing model accuracy and reliability.

In summary, the near-surface model produced through SJI not only improves seismic imaging and interpretation but also serves as an optimal starting point for advanced inversion workflows. By leveraging a broader spectrum of the seismic wavefield and integrating multiple data domains, SJI enables a more accurate, resilient, and geologically meaningful understanding of the shallow subsurface.

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